

Roamly Guide

Explore, Contribute, Unlock

By An Nguyen

Repository & How to Run

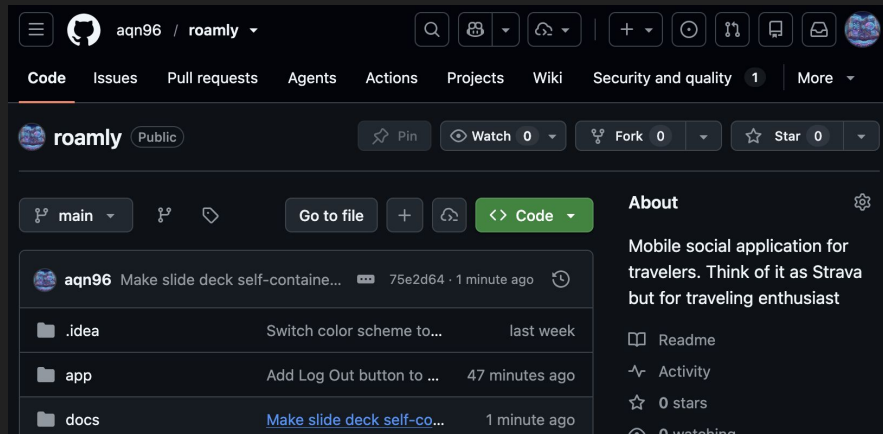
- GitHub:
<https://github.com/aqn96/roamly>
- Firebase config (google-services.json) is included in the repo — runs out of the box.
- Demo account

Email: grader@roamly.app

Password: RoamlyDemo1

(or sign up your own)

- Tested on: Pixel 8 emulator · Android 16 (API 36) · arm64-v8a · min SDK 28



System Architecture



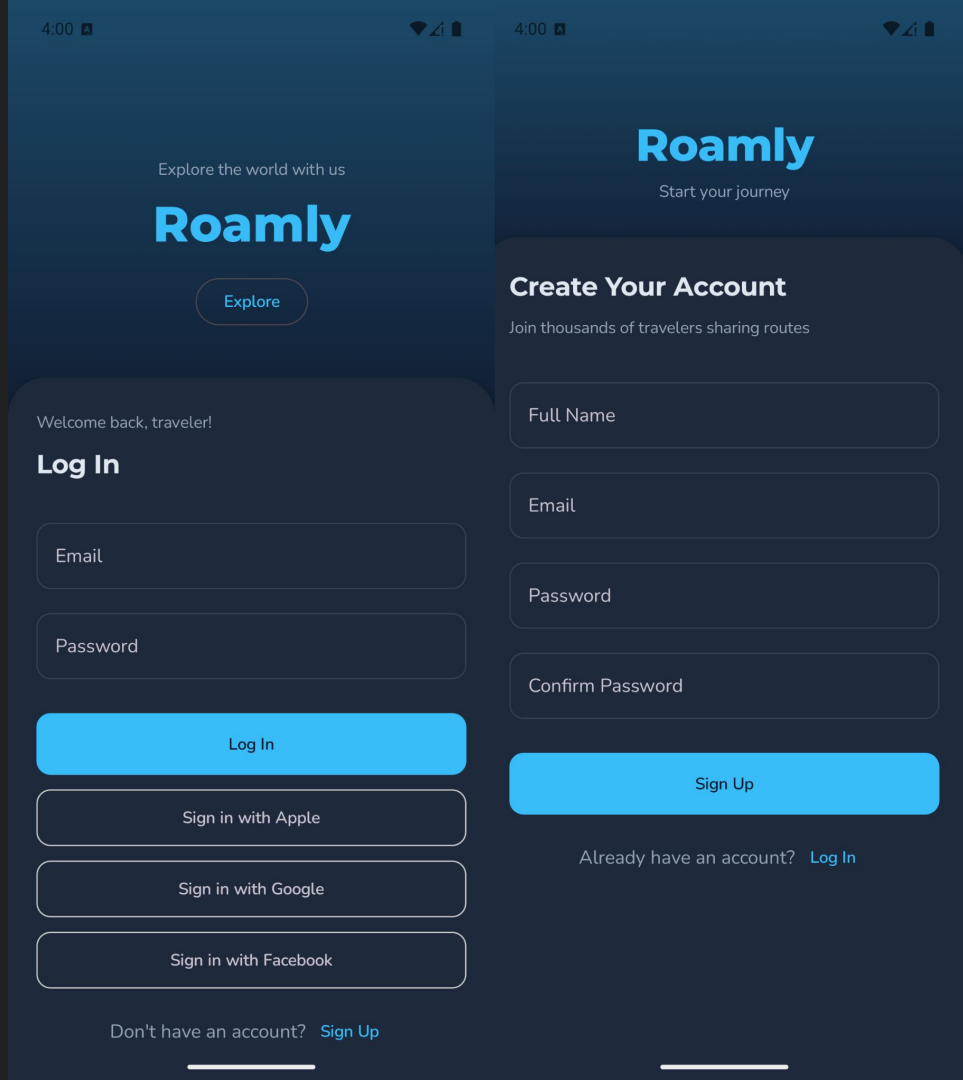
Multi-Screen Navigation

- 11 screens wired through a single Navigation Compose graph
- Screens:
Login · Sign Up · Create Profile
· Home · Active Trip · Trip Summary · Discover · Post Detail · Profile · Favorites · Location Permission
- Bottom navigation bar (Home / Discover / Favorites) preserves each tab's state



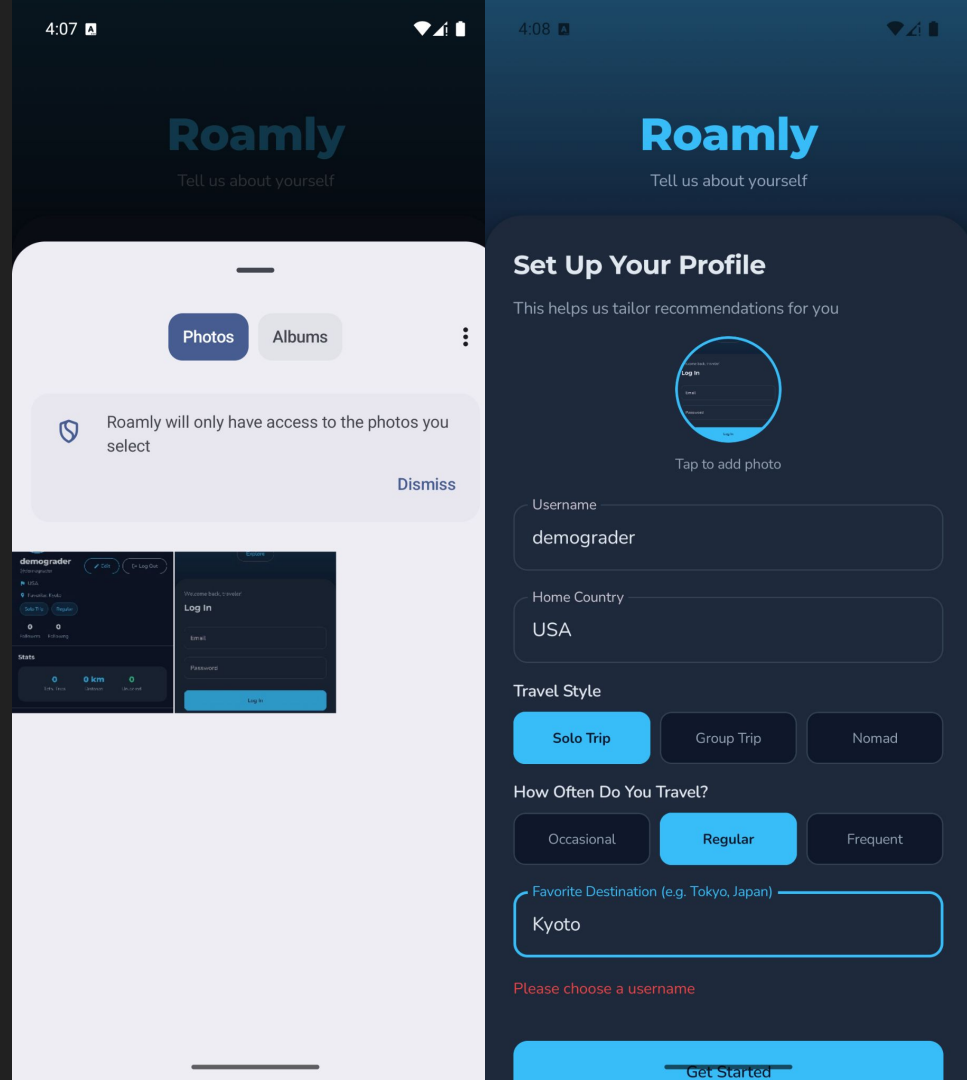
Authentication

- Email/Password sign-up and login via Firebase Authentication
- Input validation + loading states handled in the AuthViewModel
- Session persists and reopening the app skips straight to Home



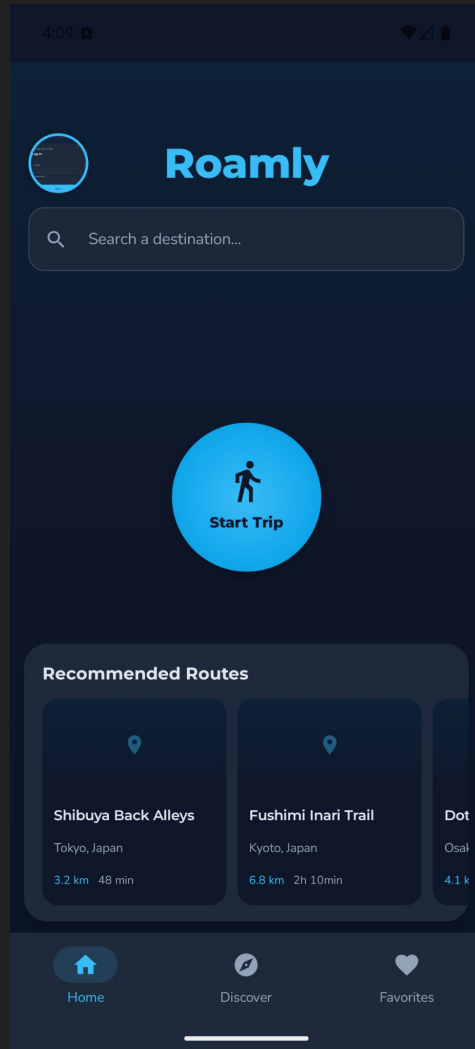
Create Profile + Photo Upload (Firebase Storage)

- Pick a profile photo from gallery/android phone
- Image uploads to Firebase Cloud Storage at profile_photos/{uid}
- Download URL is saved to the user's Firestore document and the photo shows everywhere



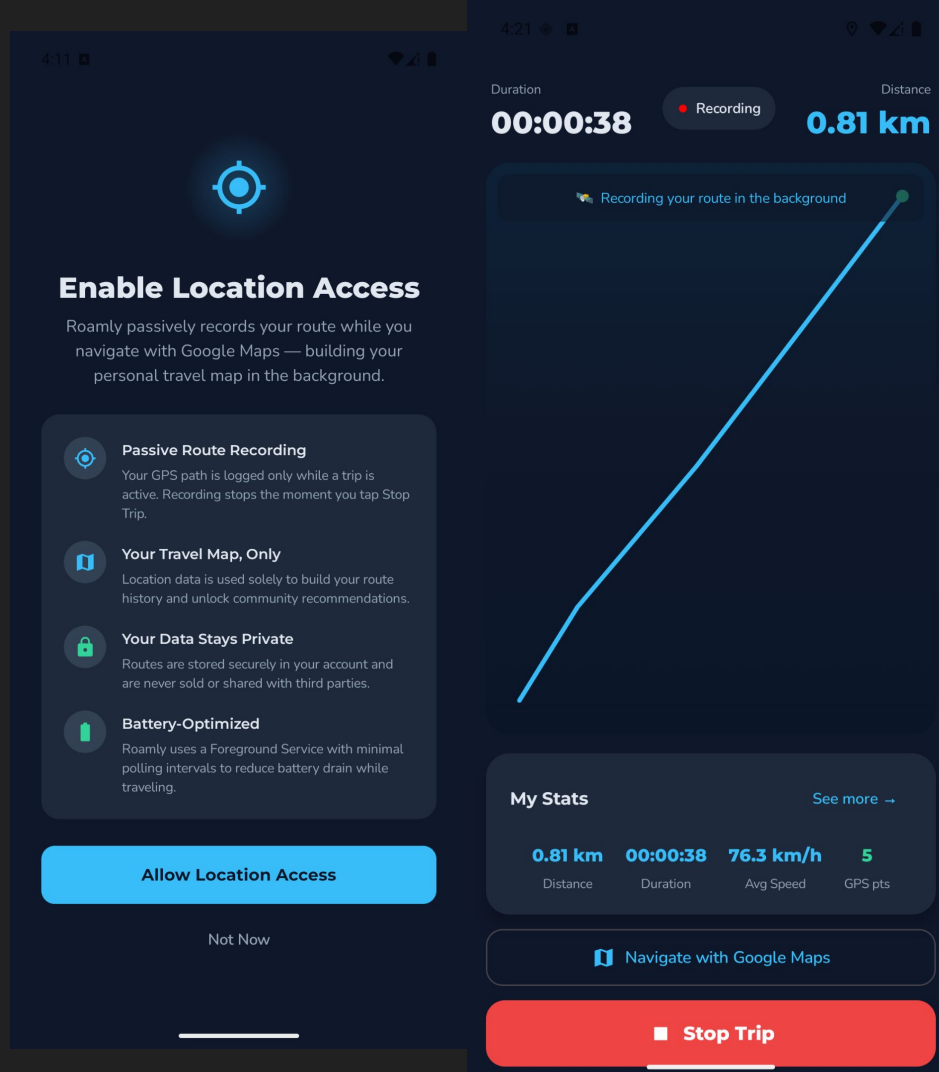
Home Page

- profile avatar (loaded from Storage) with search bar, Start Trip, recommended routes
- Simplistic UI so users can search location and go for easy tracking



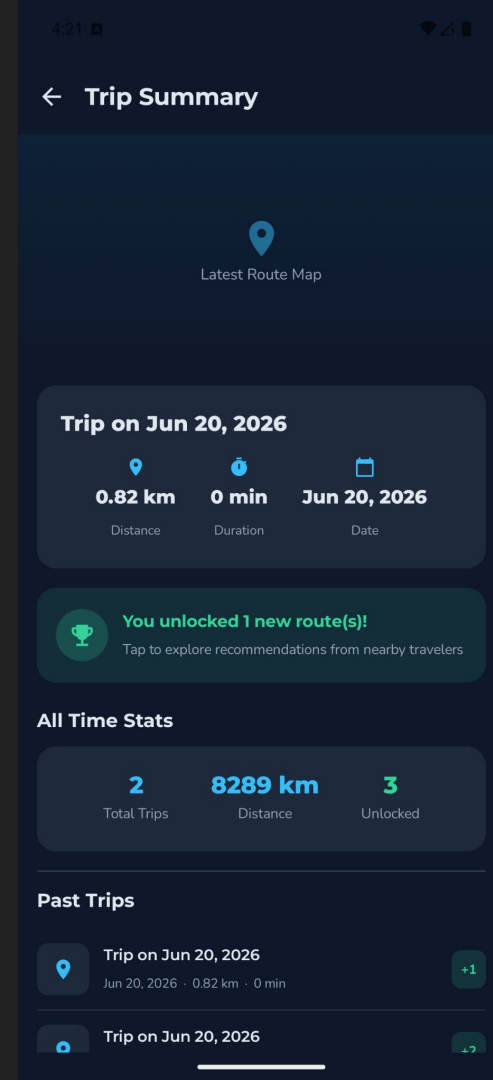
GPS Trip Recording (sensor + foreground service)

- Foreground Service + FusedLocationProvider logs the GPS path with an ongoing notification
- Route drawn live on a Compose Canvas with real-time distance, duration, and speed (DEMO)
- Navigate with Google Maps hands off navigation via Intent (Implemented but not in DEMO)



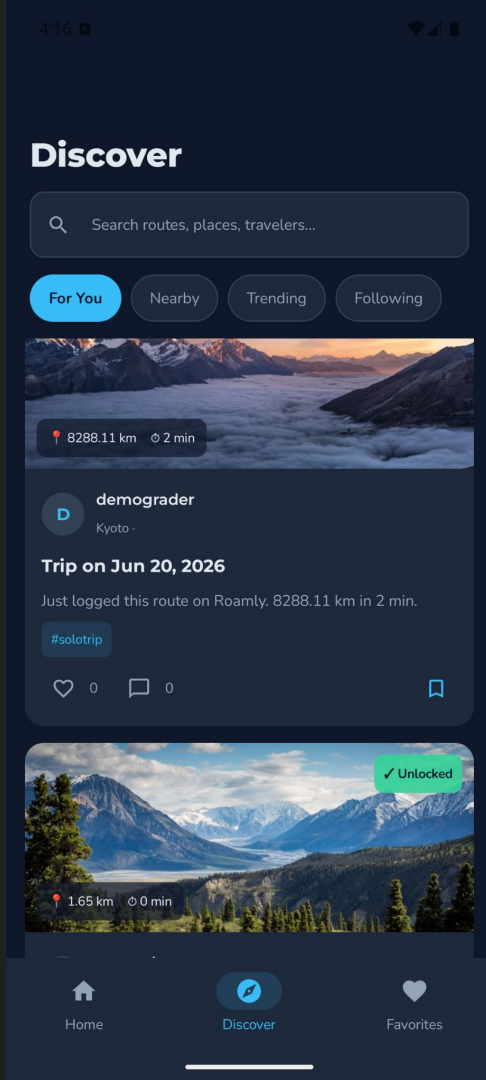
Trip Summary → Auto-Published Post

- On Stop: trip is written to Cloud Firestore and all-time stats update
- The trip is automatically published as a public post and this is the Contribution half of the app loop



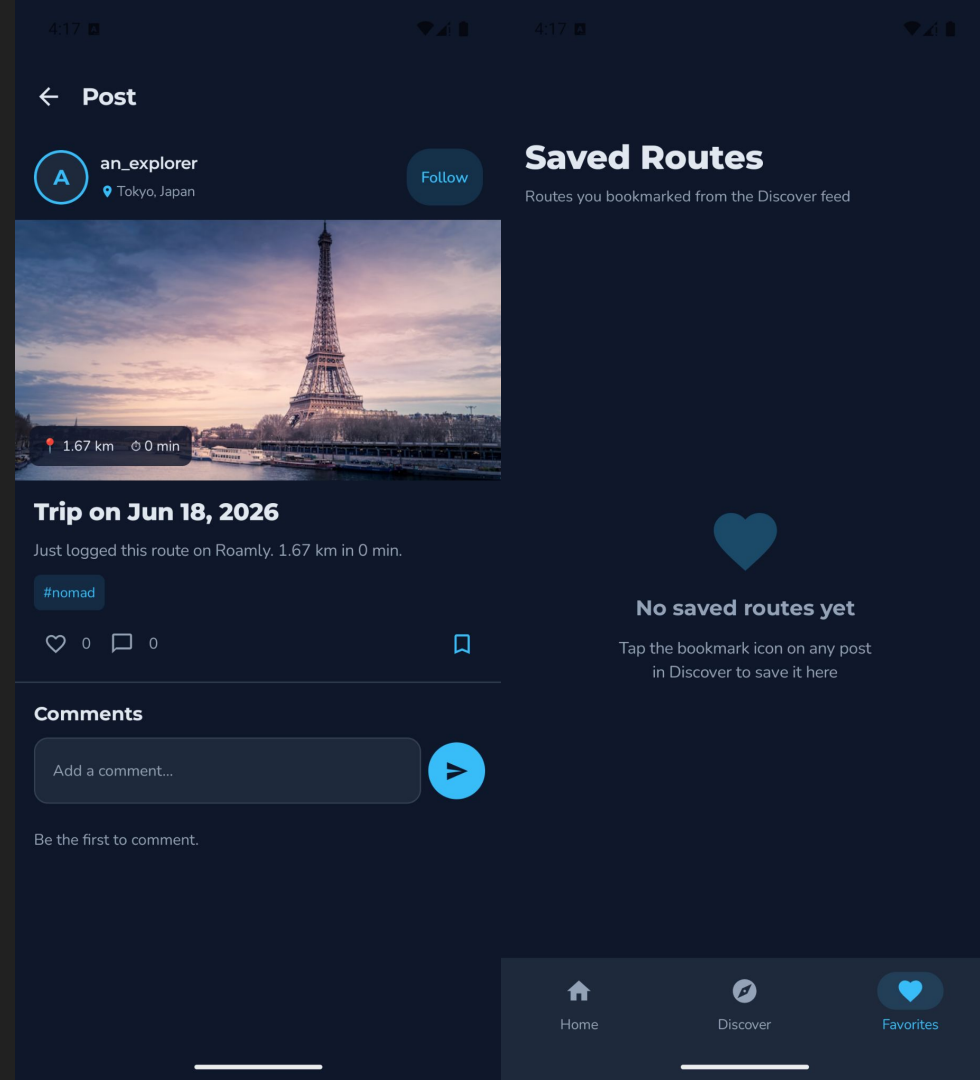
Discover Feed

- Social feed of community routes read from Firestore into a lazy scrollable list
- Filter chips: For You / Nearby / Trending / Following · like & comment counts per card



Post Detail: Like & Comment + Favorites

- Open a post to see its detail, likes, and comment thread
- Liking and commenting write to Firestore and update counts in real time
- Bookmark any route; favorites stored per-user in Firestore and shown on the Favorites tab



Your Profile (stats, edit, log out)

- Own profile: photo, real stats, social counts, computed travel level
- Edit profile, or Log Out — clears the session and returns to Login (switch/create accounts)



Multi-User: Other Profiles & Follow

- Tap any traveler to view their profile with real stats and a Follow button
- Follow/unfollow updates both users' follower/following counts in Firestore
- "Travelers to Follow" row suggests new people



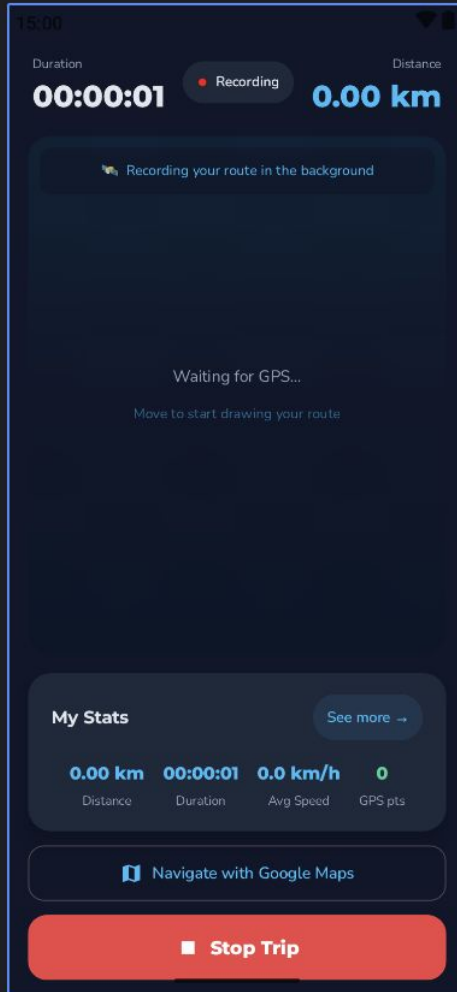
Cloud Backend Proof

- Firestore collections: users, posts, comments, favorites, trips, follow data
- Cloud Storage bucket: profile_photos/ holding uploaded images
- Real cloud storage and retrieval across the whole app

Code Quality & Documentation

- What/Who/When header comment on every source file
- @Preview on every screen & component
- MVVM + StateFlow · 60+ meaningful commits

```
1 package com.roamly.app
2
3 import android.os.Bundle
4 import androidx.activity.ComponentActivity
5 import androidx.activity.compose.setContent
6 import androidx.activity.enableEdgeToEdge
7 import com.roamly.app.navigation.RoamlyNavGraph
8 import com.roamly.app.ui.theme.RoamlyTheme
9
10 /**
11  * What: The single Activity that hosts Roamly's Compose UI. It applies the Midnight Nomad
12  *       theme and launches the type-safe navigation graph (RoamlyNavGraph), which owns every
13  *       screen and transition.
14  * Who:  An Nguyen
15  * When: Goal 7 - Final project (Jun 2026)
16  */
17 @Usage
18 class MainActivity : ComponentActivity() {
19     override fun onCreate(savedInstanceState: Bundle?) {
20         super.onCreate(savedInstanceState)
21         enableEdgeToEdge()
22         setContent {
```



Thank You!

In Summary:

Record a trip → contribute it →
unlock community
recommendations

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LinkedIn:

<https://www.linkedin.com/in/an-q-n-guyen/>

Repo:

<https://github.com/aqn96/roamly>

